

PEI-YU LIN

🏠Aachen, Germany — ✉peggy2035344@gmail.com — inboboloiono — 🌐boboloiono — ☎+49 15227202463

EDUCATION

RWTH Aachen University, Aachen, Germany

Apr 2024 – Present

M.Sc. in Electrical Engineering

- Courses: VLSI for DSP (1) (2), FPGA, EDA, HPC, Microprocessor, Compiler, Computer Arithmetic (note: 1.0)
- **Master Thesis @ETH Zurich**: Integration of Outer-product Engine into a RISC-V Vector Processor & Sparsity Exploration.
- Semester Seminar: "An Arm SME-like Vector-Matrix Extension (VME) for RISC-V with Dedicated Matrix Register Files"
- Semester Project: Embedded processor design and optimization for SHA256 algorithms via Synopsys ASIP-Designer GUI

National Cheng Kung University (NCKU), Tainan, Taiwan

Sep 2019 – Jan 2024

B.Sc. in Electrical Engineering

- a Dean's List in 2022 with **GPA 4.3 / 4.3**; Young College Elite; National Golden Prize for startup competitions * 3
- Courses: Electronics(A), Computer Organization(A), Data Structure, Logic Design, MCU, Logic Design, **Digital IC Design**
- Clubs: president of Entrepreneur Association; member of Mountaineering Association; IEEE student member

WORK EXPERIENCE

HW-SW Co-Design Intern

(7 months) Sep 2025 – Mar 2026

imec, Interuniversity Microelectronics Centre

Leuven, Belgium

Topic: Control Plane development RISC-V System-level Integration on digital AI accelerator

- Implement a Distributed Control Plane for a **multi-thread VLIW architecture** by introducing a novel instruction memory organization, including L0 instruction buffers, global/local controllers and the instruction-bundle synchronization mechanisms.
- Integrating this Soft-SIMD based tile accelerator on the RISC-V X-HEEP MCU using DMA and MMIO bus configuration
- Weekly collaborate with PhD candidates in EPFL, KTH, KU Leuven

Research Assistant

(10 months) Oct 2024 – Jul 2025

Chair of High Frequency Electronics

RWTH Aachen, Germany

- Configured and validated high-speed data transmission to 8 antennas channels via PLL & DAC peripherals on FD-DAC PCB.
- Designed a SPI hierarchical system and built a **FPGA library** and using **SystemVerilog** based on AMD Xilinx Virtex-7 board
- Designed and implemented a **serial interface controller** to control the logic inside the Sirion **tape-out** in August 2025.
- Simulated code and synthesized/pnr netlists via Xcelium Simulator, performed a **digital flow**, timing analysis, I/O planning

Digital IC Design Intern

Jul 2023 – Aug 2023

FocalTech Systems

Shenzhen, China

- Developed **SRAM read/write controller** using eXternal Serial Interface with single & burst data transfer modes.
- Created detailed command lists, state diagrams, and waveforms for SPI operations using Microsoft Visio.
- Resolved **cross-clock domain** challenges by designing synchronizers and dual-clock FIFOs for metastability mitigation.

PROJECT

VLSI Design Lab

Oct 2024 – Feb 2025

- Designed a **D flip-flop** standard cell using *Cadence Virtuoso* with ASAP 7nm PDK
- Applied Euler's Path Algorithm for layout. DRC, LVS, and PEX checks with *Calibre*; Energy simulation with *Spectre*.

Kernel Design on Memory-Mapped Near Data Processor

Jul 2025 – Aug 2025

Research Intern

Pohang University of Science and Technology, South Korea

- Designed RISC-V V based NDP kernel for parallelizable and memory-bound workloads on Docker
- Optimized GEMV, CSR-based SPMV, SSSP via Parallel Computing strategy by SIMD programming

Digital System Design for Maze Runner

GPA: 4.0

Sep 2023 – Dec 2023

📺 *Exchange Student*

University of Wisconsin-Madison, USA

- Designed and simulated key blocks like UART, PID, and SPI using Verilog with self-checking testbenches in **ModelSim**.
- Developed synthesis scripts with **DC shell** for timing, PVT, and parasitic capacitance using *Synopsys 32nm edu library*.
- Verified synthesized netlists via post-synthesis simulation and gained knowledge **IC Compiler** for APR implementation.

SKILLS

- **Programming**: (System-)Verilog · C · TCL · Perl · Linux · Python · C++ · Matlab · Git
- **Software**: Cadence EDA · Xcelium, Innovus, Virtuoso · Synopsys EDA · VCS, DC, PrimeTime, ICC2 · Siemens EDA · ModelSim, Questa, Calibre (LVS, DRC) · AMD Xilinx Vivado · Microsoft Visio · SPICE · Altium Designer
- **Languages**: Chinese: Native, English: Professional Proficiency, German: Beginner
- **Hobbys**: Travel Photographer 📷, Internet Content Writer 📝